

Qatar-1b

R-0028

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_260505 · created 2026-07-08T14:43:08+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461165.8523 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.056
Transit depth (Rp/Rs)^2	3.65 +/- 2.15 [%]
Orbital Inclination (inc)	85.35 +/- 2.81
Airmass coefficient 1 (a1)	1.98 +/- 0.94
Airmass coefficient 2 (a2)	-0.47 +/- 0.38
Scatter in the residuals of the lightcurve fit is	44.87 %
Best Comparison Star	None
Optimal Aperture	4.87
Optimal Annulus	22.7
Transit Duration (day)	0.073 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.056 vs 0.1463 (deviation 0.8σ)
Duration fit vs published	0.073 d vs 0.0692 d (deviation 0.23σ)
Mid-time O–C	6.2 min
Depth SNR	3.4
Residual scatter	44.87 %

Light curve

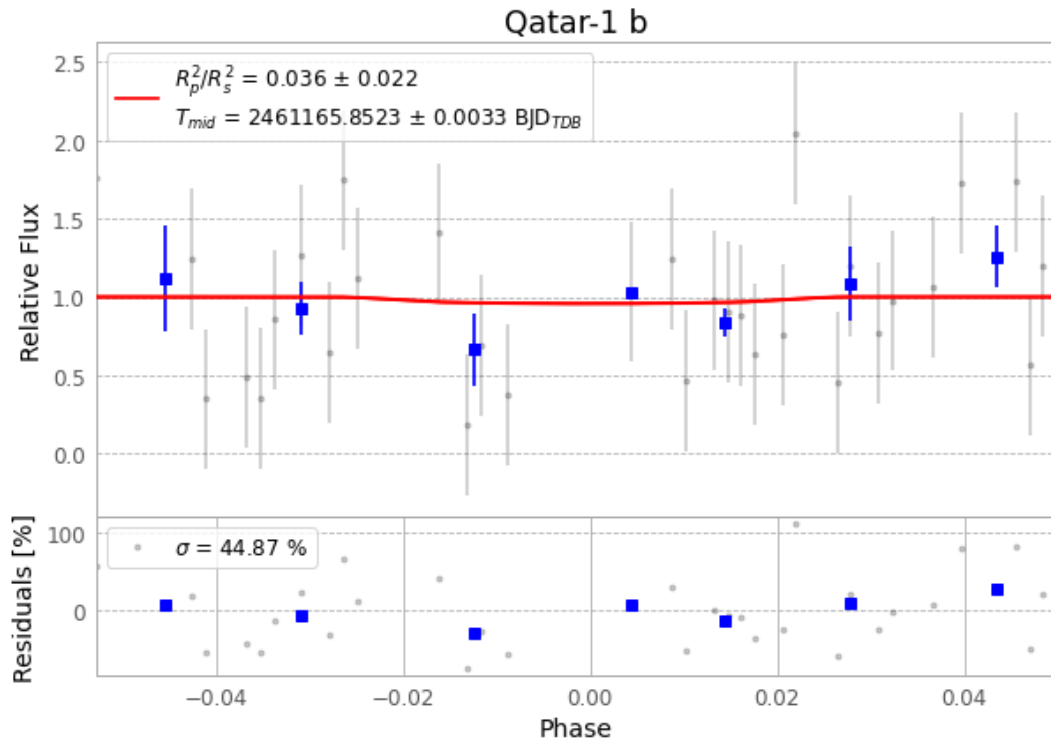


Figure 1 — FinalLightCurve_Qatar-1 b_2026-05-05.png (SHA-256 2dd181a7cfa9149c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [430, 166], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df9935af306c90fa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

74 FITS frames (49 MB), Qatar-1260505063316.FITS → Qatar-1260505101217.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260505.log (8d3f6760a3f1...), FinalLightCurve_Qatar-1 b_2026-05-05.png (2dd181a7cfa9...), FinalLightCurve_Qatar-1 b_2026-05-05.csv (ecca3193a741...)

Compute resources

Wall time 156.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.